Adnan Armouti

New York, NY | aa2546@cornell.edu | adnan-armouti.github.io | Google Scholar | GitHub

Interests: Physical AI, RF Spatial Intelligence, Differentiable Rendering, Inverse Problems, Radar

Education

Ph.D., Computer Science, Cornell University — Aug 2023–May 2028 (expected)
Advisor: Rajalakshmi Nandakumar

M.S., Computer Science, Cornell University — Aug 2023–May 2026
A exam: "Enabling Large-Scale Datasets for mmWave Radar via Differentiable Rendering"

M.S., Electrical and Computer Engineering, UCLA — Sep 2021–Jun 2023
Thesis: Fusing Light and Radio Signals for More Accurate and Equitable Blood Flow Estimation

B.S., Electrical Engineering, UCLA — Sep 2017–Jun 2021

B.A., Economics, UCLA — Sep 2017–Jun 2021

Current Research

Physics-Based Differentiable Rendering and Controllable Generation for mmWave Radar
Committee: Rajalakshmi Nandakumar (Chair), Andrew Owens, Steve Marschner, Alyosha Molnar

- I develop explicit, physics-based differentiable renderers to solve radar inverse problems [P5–P7]; editing the fitted scene or sensor enables 2D-to-3D lifting/super-resolution [P5], novel views [P6], and novel motions [P7].
- Custom CUDA kernels make dataset-scale inverse rendering and billions of synthetic chirps tractable [P7].

Publications

Selected Papers — * equal contribution

[P7] **Adnan Armouti**, Guanyue Qian, Rajalakshmi Nandakumar. "4D Point Splatting for mmWave Radar Micro-Doppler Differentiable Rendering." *Under review.*

[P6] **Adnan Armouti**, Yixuan Gao, Rajalakshmi Nandakumar. "3D Point Splatting for mmWave Radar Novel View Synthesis." *NeurIPS 2026.*

[P5] **Adnan Armouti**, Yixuan Gao, Rajalakshmi Nandakumar. "mmIR: Frequency-Space Inverse Rendering for 3D Millimeter-Wave Radar ADC Synthesis." *ECCV 2026.*

[P4] Alexander Vilesov*, Pradyumna Chari*, **Adnan Armouti***, Anirudh Bindiganavale Harish, Kimaya Kulkarni, Ananya Deoghare, Laleh Jalilian, Achuta Kadambi. "Blending Camera and 77 GHz Radar Sensing for Equitable, Robust Plethysmography." *ACM Transactions on Graphics (SIGGRAPH 2022).*

[P3] Tanvir Ahmed, Yixuan Gao, **Adnan Armouti**, Rajalakshmi Nandakumar. "mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption." *MobiCom 2026.*

[P2] Pradyumna Chari*, Anirudh Bindiganavale Harish*, **Adnan Armouti**, Alexander Vilesov, Sanjit Sarda, Laleh Jalilian, Achuta Kadambi. "Implicit Neural Models to Extract Heart Rate from Video." *ECCV 2024.*

[P1] Kai Del Regno, Alexander Vilesov, **Adnan Armouti**, Anirudh Bindiganavale Harish, Selim Emir Can, Ashley Kita, Achuta Kadambi. "Thermal Imaging and Radar for Remote Sleep Monitoring of Breathing and Apnea." *arXiv preprint (2024).*

Patents

[US2] A. Kadambi, A. Kita, A. Vilesov, K. Del Regno, S. E. Can, L. Jalilian, A. B. Harish, **A. Armouti**. "Methods and Apparatus to Detect and Classify Forms of Sleep Apnea." U.S. Provisional Application, 2024.

[US1] A. Kadambi, L. Jalilian, P. Chari, C. Talegaonkar, D. Karinca, M. Cannesson, K. Kabra, O. Salehi-Abari, A. Kita, **A. Armouti**. "Systems and Methods for Measuring Vital Signs Using Multimodal Health Sensing Platforms." U.S. Patent Application 2023/0233091, 2023.

Honors and Awards

NVIDIA Academic Grant — Jun 2026
UCLA ECE Outstanding Master of Science Student Award — Jun 2023
UCLA ECE Distinguished Master's Thesis Research Award — May 2023
NSF REU Fellowship — Jun 2021
Intel SRC URP Scholarship — Jun 2020

Research Experience

Wireless Systems Lab, Cornell University, New York, NY — Aug 2023–Present
PI: Rajalakshmi Nandakumar

- Built mmIR, an open-source differentiable FMCW radar inverse renderer with multi-bounce propagation, polarization, and diffraction; 0.914 RA correlation on 13 scenes vs. Sionna-RT's 0.307 [P5].
- Built 3DPS, the first differentiable point renderer for radar novel view synthesis, producing ADC, range profiles, and range–azimuth maps from one model; 1.7–5.2× over NVS baselines [P6].
- Built 4DPS, which extends point splatting to moving scenes through Doppler; a pose model trained only on its renders reaches 21.9 cm MPJPE on real data, vs. 127.6 cm for RF-Genesis [P7].

Visual Machines Group, UCLA — Sep 2020–Jun 2023
PI: Achuta Kadambi

- Co-led a successful $1M DARPA grant proposal funding camera–radar fusion for equitable plethysmography [P4].
- Co-implemented a neural implicit decomposition for heart-rate estimation on out-of-distribution subjects [P2].
- Co-implemented a thermal and radar fusion method for remote sleep-apnea classification and monitoring [P1].

Internships

UCLA Health, Sleep Disorders Center, Los Angeles, CA — Jun 2022–Sep 2022
Summer Research Intern — *Advisor:* Ashley Kita

UCLA Visual Machines Group, Los Angeles, CA — Jun 2021–Sep 2021
Summer Research Intern — *Advisor:* Achuta Kadambi

Open-Source Contributions

4DPS (project page). Differentiable point splatting for moving scenes and micro-Doppler [P7].
3DPS (GitHub). Differentiable point splatting for radar novel view synthesis [P6].
mmIR (GitHub). Differentiable FMCW radar inverse renderer [P5].
EquiPleth (GitHub). Camera–radar fusion for equitable plethysmography [P4].
FastImplicitPleth (GitHub). Implicit neural models for heart rate from video [P2].
NonContactApneaDetection (GitHub). Thermal and radar sleep-apnea detection [P1].

Public Demonstrations

Tanvir Ahmed, Yixuan Gao, **Adnan Armouti**, Rajalakshmi Nandakumar. "mmFHE: mmWave Sensing with End-to-End Fully Homomorphic Encryption." *MobiCom Demos 2026* [P3].

Media Coverage

- EquiPleth [P4]: Forbes (Jan 2023), Daily Bruin (Sep 2022), UCLA Newsroom (Aug 2022)

Teaching

Teaching Assistant, Cornell Tech, Cornell University — Aug 2023–Present

- CS 5785 "Applied Machine Learning," Prof. Kyra Gan (Fall 2023, Fall 2024, Fall 2025, Fall 2026)
- CS 5788 "Introduction to Generative Models," Prof. Andrew Owens (Spring 2026)
- INFO 5368 "Practical Applications in Machine Learning," Prof. Angelique Taylor (Spring 2025, Spring 2026)

Mentorship

Guanyue (Thomas) Qian, *B.S. Student, NYU*	Jun 2026–Present
Kai Del Regno, *M.S. Student, UCLA*	Sep 2022–Jun 2023
Jianchong (Devin) Ma, *M.S. Student, Stanford University*	Sep 2022–Jun 2023
Zoe (Rui) Ma, *M.S. Student, Columbia University*	Sep 2022–Jun 2023

Referee Service

ICLR	2027–
NeurIPS	2026–

Skills

- **ML / Computer Vision / Graphics**: PyTorch, JAX, TensorFlow, Mitsuba 3, Dr.Jit, Slang, CUDA, OptiX
- **Programming & Tools**: Python, C/C++, MATLAB; Git, Linux, Docker, LaTeX

References

Prof. Rajalakshmi Nandakumar, Cornell University, rn283@cornell.edu
Prof. Andrew Owens, Cornell University, andrew.owens@cornell.edu
Prof. Steve Marschner, Cornell University, srm@cs.cornell.edu
Prof. Kyra Gan, Cornell University, jg2424@cornell.edu
Prof. Achuta Kadambi, UCLA, achuta@ee.ucla.edu